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ORIGINAL RESEARCH ARTICLE

Restored Oxbow Wetlands within an Agricultural Landscape: Using Physical and Biological Characteristics to Evaluate Impacts of Tile Drainage Input

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Abstract
Oxbow wetlands have been restored in the Midwestern United States to provide habitat for wetland-dependent species and to sequester contaminants originating from agricultural activities. Intensive agriculture may have adverse impacts on oxbow functions, especially if wetlands receive water inputs from subsurface drainage systems (e.g., tile drainage). To explore the influence of tile drainage on oxbow wetland communities, we quantified relationships between physical and biotic variables in 12 Iowa, USA oxbows over a two-year period (June to August 2019–2020). Six oxbows received direct water inputs from tile drainage (multipurpose oxbows), whereas remaining sites did not (non-tiled oxbows). In each oxbow, we measured physical variables and documented taxonomic composition, diversity, and abundance of macrophytes, macroinvertebrates, and fishes. Although water temperature was lower in multipurpose oxbows, values for other physical variables (e.g., turbidity, conductivity, and total dissolved solids) were similar across sites. No significant difference was detected for any biotic variable between oxbow types. In total, we observed 44 invertebrate taxa in both oxbow types with an average richness of 18.6 in non-tiled oxbows and 17.5 in tile-fed oxbows. We sampled 35 fish species, with an average richness across sampling dates of 8.2 in non-tiled oxbows and 11.4 in multipurpose oxbows. A total of 2682 Topeka shiners were found in both non-tiled and multipurpose oxbows. Non-metric multidimensional scaling revealed that potential physical determinants of macrophyte, invertebrate, and fish abundance were unrelated to tile drainage. Tile drainage had negligible impacts on coarse physical characteristics, taxa richness (fish and macroinvertebrates) and abundance (fish and macroinvertebrates).

Keywords Wetland Ecology · Conservation Biology · Fisheries Biology · Habitat Restoration · Wetland Restoration

Introduction

In the United States, many rivers have been largely disconnected from floodplains as a result of clear-cutting, alluviation, and channelization (Walter and Merritts 2008; Blann et al. 2009; Cluer and Thorne 2014). The lack of connection, coupled with purposeful draining, has eliminated approximately 48% of the nation's wetlands with subsequent declines in water quality and biodiversity (Camargo and Alonso 2006; Davidson 2014; Fluet-Chouinard et al. 2023). These declines are particularly prevalent in areas of intensive agriculture (Camargo and Alonso 2006; Cluer and Thorne 2014).

The upper Mississippi River Valley is emblematic of the shift to agricultural landscapes. Historically, local river ecosystems were characterized by meandering channels with diverse habitat structure and abundant off-channel depressional wetlands, including oxbow wetlands (Menzel 1981; Blann et al. 2009). Oxbow wetlands (hereafter referred to as oxbows) form when river meanders become disconnected from the main channel through natural erosional processes (Rowland et al. 2005; Ishii and Hori 2016). Like other riverine wetlands, oxbows improve floodplain ecosystem services by intercepting and storing water and sediment, thereby reducing inputs to the adjacent stream channel while providing habitat for many wetland-dependent species (Naus and Adams 2018; Schilling et al. 2018a). One such species is the Topeka shiner (*Notropis topeka*), an endangered fish

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generally restricted to shallow slow-moving stream pools and off-channel habitat, including oxbows (Kerns and Bonneau 2002; Sindt et al. 2012; Pierce et al. 2019).

As streams were increasingly channelized throughout the Midwestern agricultural landscape, oxbow formation was largely eliminated, and many existing wetlands, both natural and those created through channelization, were drained and filled with sediment (Schneiders 1996; Rhemtulla et al. 2007; Blann et al. 2009; Cluer and Thorne 2014). These losses, in addition to increased application of herbicides, pesticides, and fertilizers, contributed to degraded water quality (Camargo and Alonso 2006), and declines in species of conservation concern (Tabor 1998; Bakevich 2015). In the upper Mississippi River Valley today, most oxbows that remain on the landscape were created through channelization, and are disappearing over time through sedimentation (Rhemtulla et al. 2007; Blann et al. 2009).

Beginning in 2002, oxbows have been restored in Iowa to retain floodwater, sequester nutrients, and provide habitat for Topeka shiner and other biota (Kenney 2018; Schilling et al. 2018b, 2019). Typically, restorations involve excavation to reestablish a groundwater connection, allowing the resulting depressions to retain standing water (Kenney 2018). Restored oxbows can sequester considerable quantities of nutrients, particularly when receiving water from agricultural drainage systems, including underground networks of pipes (i.e., tile drainage; Schilling et al. 2017, 2018b; Pierce and Schilling 2023). Additionally, multiple studies have found that these restored oxbows harbor a high diversity of fish species (Kenney 2018; Simpson et al. 2019; Pierce et al. 2019; Osterhaus 2021). Surveys suggest that the increased restoration of oxbows has contributed to a recent reversal of Topeka shiner declines in Iowa and Minnesota streams, with a 25% increase in population size between 2010 and 2017 (Bakevich 2015; Pierce et al. 2019; Osterhaus et al. 2021).

Oxbows designed to simultaneously intercept tile drainage water and provide habitat (hereafter multipurpose oxbows) were recently recognized as an approved nutrient export reduction practice by the Science Advisory Team of the Iowa Nutrient Research Center (INRC) (Schilling et al. 2019). However, the relationship between tile drainage inputs and both habitat and biotic community characteristics in oxbows has not been thoroughly studied, with previous oxbow biotic community surveys generally restricted to one sampling date (Bakevich 2015; Simpson et al. 2019; Osterhaus et al. 2022a). Given the increased pesticide (Kraus et al. 2021; Campbell et al. 2023), selenium (Henry et al. 2020), and nutrient loads (Pierce and Schilling 2023), associated with tile water, it is possible that conditions within multipurpose oxbows may be harmful to the ecology of these systems.

Our overall objective was to characterize physical and biological effects of tile drainage input on restored oxbow wetlands. Physical and biological attributes may be affected by tile drainage in a number of ways and are themselves interrelated. We predicted that oxbows would be dominated by a few taxa tolerant of agricultural contaminants (Obolewski et al. 2016; Cavallaro et al. 2019). Because tile drainage is typically higher in nitrate and orthophosphate (PO4) than local groundwater, nutrient introductions from tile drainage will likely contribute to eutrophication (Smith 2003; Blann et al. 2009; Schilling et al. 2018c), and we hypothesized that multipurpose oxbows would therefore be higher in temperature and turbidity, as groundwater is typically cooler than surface water and high biomass of planktonic and floating algae increases suspended particle concentrations and absorbs heat (Paaijmans et al. 2008; Bayley et al. 2013). We expected that macrophyte diversity would be lower in more eutrophic multipurpose wetlands, as the decreased depth of light penetration would inhibit the growth and establishment of submerged aquatic vegetation (Lougheed et al. 2001; Bayley et al. 2013; Maurer et al. 2014). Given that eutrophication of wetlands is associated with decreased invertebrate and fish diversity (Hall and Rudstam 1999; Jeppesen et al. 2000; Spieles and Mitsch 2000; Smith 2003; Tarr et al. 2005; Menetrey et al. 2008), we predicted that tile-fed oxbows would have lower diversity of fishes and invertebrates than non-tiled oxbows. We hypothesized two pathways by which tile drainage could affect oxbow ecology. The first is directly through changes in physical variables (e.g., an increase in nutrient levels reduces macrophyte taxonomic diversity) and the second is indirectly through cascading effects on the biotic community (e.g., a less diverse macrophyte assemblage supports fewer macroinvertebrates; Fig. 1; Table 1).

Methods

Study sites

Twelve oxbows were included in the study, six of which were constructed to receive subsurface tile drainage (multipurpose oxbows), and six oxbows that did not have tile inputs (hereafter, non-tiled oxbows). Oxbow surface area ranged from 209 to 2440 m^2, with an average surface area of 916 m^2 and volume also varied considerably, ranging from 78–1779 m^3 (Table 2). All oxbows were restored between 2017 and 2019. Oxbows were located in the Boone HUC 8 (10 oxbows) and North Raccoon HUC 8 (2 oxbows) watersheds of north central and west central Iowa, respectively (Table 2; Pierce et al. 2019). Total agricultural surface cover of the receiving catchment area of the oxbows ranged between 85.9%-94.7% (Table 2; Dewitz 2014). During the summers of 2019 and 2020 we collected data for several physical variables in addition to abundance and taxonomic

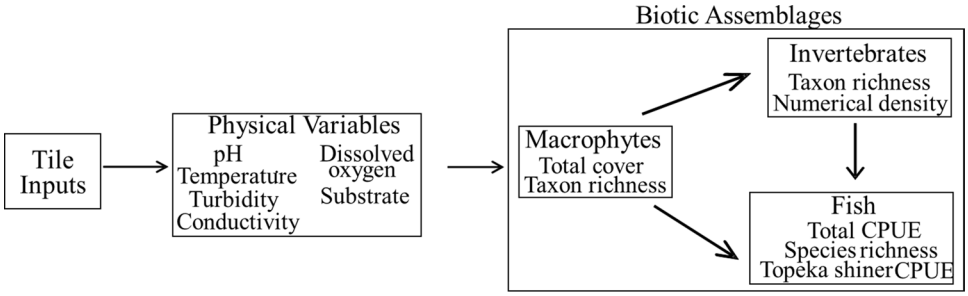


Fig. 1 Hypothesized pathways of tile drainage effects on the biotic community (macrophytes, invertebrates, and fish) of oxbows directly through changes in the physical variables (substrate, pH, turbidity, dissolved oxygen, and temperature), and indirectly through other changes in the biotic community

Table 1 Physical and biological variables recorded in oxbow wetlands. Fish catch per unit effort (CPUE) refers to the number of individuals per $100m^2$

Variable Type	Variables measured
Physical	Surface area, depth, substrate, canopy cover, temperature, turbidity, TDS (total dissolved solids) ORP (oxidation reduction potential), pH, dissolved oxygen, and conductivity
Biological	Macrophyte assemblage (taxon richness, cover by taxa, and total cover), Macroinvertebrate assemblage (numerical density and taxon richness) Fish assemblage (total CPUE, species richness, and Topeka shiner CPUE)

Table 2 Characteristics of each oxbow site sampled for this study. Percent agricultural land is defined as the total area of rangeland and cropland in the receiving watershed of each oxbow (Dewitz 2014). Flow point data were unavailable for site 6 and the receiving watershed could not be defined

Site	HUC 8	Date of Restoration	Oxbow Type	Surface Area (m^2)	Volume (m^3)	Percent Agricultural Land	Monthly Precipitation (inches)
1	Boone	January 2017	Non-tiled	828	340	94.1	6.33
2	Boone	January 2017	Non-tiled	568	197	94.1	6.33
3	North Raccoon	December 2017	Non-tiled	1981	1779	90.3	5.57
4	North Raccoon	December 2017	Non-tiled	366	139	85.9	5.37
5	Boone	April 2019	Non-tiled	510	237	94.6	6.28
6	Boone	December 2017	Non-tiled	437	210	-	5.84
7	Boone	April 2019	Multipurpose	580	247	92.6	6.15
8	Boone	January 2017	Multipurpose	711	364	94.4	6.27
9	Boone	April 2019	Multipurpose	222	78	94.7	6.28
10	Boone	April 2019	Multipurpose	1801	1377	94.7	6.28
11	Boone	January 2017	Multipurpose	807	398	92.7	6.15
12	Boone	March 2019	Multipurpose	2179	1084	92.7	6.15

diversity of fishes, macroinvertebrates, and macrophytes. We then quantified relationships between presence/absence of tile drainage inflow, physical variables, and biological variables (Table 1) to identify potential influences of tile drainage on organism abundance and taxonomic diversity.

Physical variables

We measured physical variables in each oxbow following a modified procedure of Simpson et al. (2019). Measurement of physical variables occurred prior to collecting biological community data or samples. We measured turbidity at each oxbow on four dates between late June and early August. On each date (8–16 days apart), we measured turbidity at a depth of 40 cm at three locations spaced equidistantly along the wetted length of the oxbow. The turbidity value reported for each site and date was the average value calculated from these three measurements.

We measured all other physical variables on two dates annually, once in mid-July and once in mid-August. Prior to data collection, each oxbow was divided into five transects that extended from shoreline to shoreline at 10, 30, 50, 70, and 90% of wetted length. At each transect, we obtained wetted width and recorded measurements for remaining physical variables at distances equivalent to 25, 50, and 75% wetted width for a total of 15 measurements per sampling date. At

each location, we measured overhead canopy cover using a spherical densiometer. We recorded predominant substrate type within a 0.5m^2 area and scored it on a five-point scale (silt = 1, sand = 2, gravel = 3, cobble = 4, bedrock = 5). We measured temperature, pH, oxidation reduction potential (ORP), dissolved oxygen, conductivity, resistivity, salinity, and total dissolved solids (TDS) with a HI 9829 water quality meter (Hanna Instruments). Turbidity was measured using a 2100Q portable turbidimeter (Hach). Average values for each physical variable across all sampling dates were used in statistical analyses. While averaging noisier variables (e.g. turbidity) can make detecting differences across oxbow types difficult, we did not sample frequently enough to identify meaningful trends in these variables.

Macrophyte Assemblage Sampling

We quantified assemblage attributes of macrophytes (free-floating and rooted floating-leaved, emergent, and submerged nonvascular and vascular macroscopic plants) once annually, in mid-summer (July 6–18), following methods adapted from Maurer et al. (2014). At each oxbow we first established eight linear transects spanning the wetted width of the oxbow, spaced equidistantly. Each transect was further divided into three sections of equal length, resulting in a total of 24 macrophyte sampling sections. We placed a 1.0 m^2 frame on the water surface within each section, with the specific location determined by a random number generator. Visual observation and a plant rake were used to identify macrophyte taxa and measure macrophyte cover and taxon richness. Macroscopic green algae were identified as "Chlorophyta", whereas other macrophytes were identified to genus. Macrophyte taxon richness was quantified as the total number of taxa recorded across all 24 plots. Values for macrophyte cover, quantified on a scale of 0–100%, were recorded for each taxon, and for total cover. Percent cover values for all plots in an oxbow were averaged to produce one value for each macrophyte assemblage variable.

Macroinvertebrate Assemblage Sampling

We used a stovepipe sampler (36 cm diameter) to sample macroinvertebrates, according to Maurer et al. (2014). We placed the sampling device at five evenly spaced locations along the oxbows wetted length at depths ranging from 40 to 60 cm. At each location, we first removed macrophytes and pieces of coarse particulate organic matter > 5 cm long (plants/CPOM) by hand from within the stovepipe, then used a fine-mesh net (250 µm) to collect the top 2.5 cm of sediment and sweep through the water column until 10 successive sweeps revealed no visible invertebrates. All collected material retained by a 500 µm mesh sieve was combined into a single composite sample representative of that oxbow. We preserved samples in 5% buffered formalin with Rose Bengal dye, replaced with 70% ethanol after 24 h.

Macroinvertebrate samples were processed in the laboratory. We first conducted a comprehensive search for large-bodied invertebrates by placing sample contents in a pan, scanning the entire sample with an unaided eye, and removing all invertebrates that were ≥ 0.5 cm long. We then homogenized the remaining sample contents in a pan marked with ninety 27.5 cm^2 cells. We enclosed a randomly selected cell within an acrylic frame with solid walls and open top and bottom. At the cell boundaries we cut plants/CPOM contents and transferred all cell contents to a petri dish. We examined cell contents at 10X magnification and identified and counted all invertebrates. An additional cell was then randomly selected and processed in its entirety using the same method. If we had not collected at least 500 invertebrates after processing contents of two cells, we randomly selected and processed additional cells in their entirety until at least 500 individuals were removed from the larger sample. Insects and mollusks were identified to family, while other invertebrates were identified to order (Amphipoda, Cladocera, Conchostraca, Decapoda, and Oligochaeta), subclass (Acari, Collembola, Copepoda, and Hirudinea), or class (Ostrocada). While functional feeding groups and tolerance to contaminants can vary significantly within coarser taxonomic groups, finer identifications can contribute to sample bias and are beyond the scope of this study. We calculated invertebrate taxon richness as the total number of taxa recorded during subsampling and the comprehensive large-bodied search. We calculated numerical densities of invertebrates (as individuals per 1 m^2 oxbow surface area) based on the comprehensive large-bodied search and the subsampling procedure, accounting for the percentage of the sample processed by sub-sampling (Maurer et al. 2014).

Fish Assemblage Sampling

We sampled fishes on two dates annually, in early (June 24-July 1) and late (July 25-August 15) summer (n = 4). On both sampling dates we made a single pass with a bag seine (16.8 × 1.8 m, 6.35-mm mesh or 10.7 × 1.8 m, 6.35 mm mesh, depending on oxbow width) through the entire length of the oxbow, and held fish in floating mesh pens while individuals were identified to species and counted (Simpson et al. 2019; Osterhaus 2021; Osterhaus et al. 2022a). We calculated fish catch per unit effort (CPUE) for each species as the number of individuals per 100 m^2 oxbow surface area and recorded CPUE for total catch and catch by species. We averaged CPUE across sampling dates in each oxbow.

Statistical analysis

For statistical analysis we considered each oxbow to be an independent sampling unit. We performed statistical analyses in R (R Core Development Team 2019) and considered relationships to be significant at $p \leq 0.05$. Prior to analysis we used single-variate Shapiro tests and multivariate Mardia tests to determine whether data met normality assumptions of parametric statistical tests. We arcsine-square root transformed canopy cover and log-transformed surface area, depth, pH, and fish CPUE (McCune and Grace 2002). Additionally, we evaluated pairwise relationships using correlation analysis to detect any instances of multicollinearity prior to multivariate analysis of variance (MANOVA). When multicollinearities were detected, the variable with a lower Pearson correlation with the response was removed.

We used MANOVA to determine if variables collectively differed between multipurpose- and non-tiled oxbows. We ran separate MANOVAs to evaluate the effects of oxbow type on aggregated physical and aggregated biological variables (Tables 3 and 4). We then ran ANOVAs to identify individual variables (e.g., temperature, dissolved oxygen, taxon richness, etc.) that differed significantly between multipurpose and non-tiled oxbows.

We used non-metric multidimensional scaling (NMDS) ordination to examine relationships between physical variables and abundance of macrophyte, invertebrate, and fish taxa. Separate ordinations were plotted for macrophytes, invertebrates, and fish using Bray Curtis distances and the Vegan package (Oksanen et al. 2019). We plotted potential causal variables having some relationship to organism abundance ($r^2 \geq 0.25$) as vectors.

Finally, we conducted statistical analysis to determine if Topeka shiner presence was significantly related to occurrence of other taxa. Analysis of Variance Using Distance Matrices (ADONIS) was conducted for fish, invertebrate and macrophyte assemblages for sites with Topeka shiner present and absent. Prior to conducting ADONIS, assemblage data were converted to Bray–Curtis dissimilarity matrices. In cases where ADONIS detected a significant community difference between oxbow types, we subsequently ran Similarity Percentage (SIMPER) analysis to identify which taxa were strongly associated with Topeka Shiner.

Table 3 Mean and range values for physical variables measured in oxbow wetlands of central Iowa. Data are reported for both oxbow types averaged over the two years of sampling. Ranges are based on averages by sampling date

	Multipurpose oxbows n = 6		Non-tiled oxbows n = 6	
	Mean	Range	Mean	Range
Surface area (m^2)	1097.9	509.7–1800.8	733.7	221.7–1980.7
Depth (m)	0.5	0.4–0.7	0.5	0.3–0.9
Canopy cover (percentage)	6.0	0.3–20.7	4.7	0–22.4
Temperature (℃)	23.6	20.3–26.5	26.3	24.1–30.0
Turbidity (NTU)	52.7	22.0–95.6	40.7	5.8–64.8
Dissolved oxygen (mg/L)	5.2	3.0–7.0	5.6	2.4–8.3
pH	7.9	7.4–8.3	8.1	7.5–8.3
Conductivity (µs/cm)	513.5	451–530.2	455.6	354.4–541.9
ORP (mV)	56.6	10.0–86.3	44.7	-31.8–89.2
TDS (ppm)	253.5		241.1	204.1–284.6
Average substrate score	1.1	1.0–1.2	1.1	1.0–1.2

Results

Physical characteristics of multipurpose and non-tiled oxbows

Collectively, values for physical variables were not significantly different in multipurpose and non-tiled oxbows (MANOVA; Pillai Statistic = 0.989, Df = 1,10, F = 8.86, p = 0.26; Table 3). Water temperature was somewhat lower in multipurpose sites (mean = 23.6 °C, range = 20.3–26.5) than in non-tiled oxbows (mean = 26.3 °C, range = 24.1–30.0),

Table 4 Mean and ranges for all community variables in our study. Data are reported for both oxbow types averaged over the two years of sampling. Catch per unit effort refers to the number of individuals per 100 m^2 of oxbow surface area

	Multipurpose oxbows n = 6		Non-tiled oxbows n = 6	
	Mean	Range	Mean	Range
Macrophyte taxon richness	5.9	2.5–9.0	5.5	2.5–8.5
Macrophyte cover (percentage)	36.1	4.2–94.7	33.1	4.6–93.4
Invertebrate taxon richness	17.5	15.0–19.5	18.6	15.0–23.0
Invertebrate numerical density	7964.6	2656.5–18,188.7	6672.0	1334.7–14,778.5
Fish species richness	11.42	4.0–14.5	8.2	1.5–17.0
Fish CPUE	362.9	24.1–799.4	445.4	214.8–1055.3
Topeka shiner CPUE	10.3	0.0–28.2	4.9	0.0–10.6

but this difference was not statistically significant (Estimate = 2.69, t = 2.11, p = 0.06). Other measured physical habitat features did not differ between wetland types (P ≥ 0.11; Table 3).

Macrophyte assemblage

During this two-year study we recorded 27 macrophyte taxa (Tables 4 and 5). Average macrophyte taxon richness was 5.5 in non-tiled sites and 5.9 in multipurpose oxbows (ANOVA; F = 0.0083, Df = 1, 10, p = 0.78). Four genera occupying non-tiled sites were not found in multipurpose sites, while seven genera in multipurpose sites were never recorded in non-tiled oxbows (Table 5). The most common macrophyte taxon was *Potamogeton* at 24.2% average coverage in non-tiled sites and Chlorophyta (green algae) at 17.7% in multipurpose oxbows. Average total macrophyte cover was 33.1% (range = 4.6–93.4) in non-tiled oxbows and 36.1% (range = 4.2–94.7) in multipurpose sites (ANOVA; F = 0.078, Df = 1, 10, p = 0.79).

Table 5 Percent cover of plant taxa recorded in oxbow wetlands in 2019 and 2020. Taxa are listed in descending order of average cover regardless of wetland type

Taxon	Common name	Average cover by taxon (%)		
		Overall	Non-tiled oxbows	Multi-purpose oxbows
Potamogeton	Pond weeds	18.3	24.2	12.4
Chlorophyta	Green Algae	12.5	7.4	17.7
Lemna	Duckweeds	7.3	0.6	14.1
Phalaris	Canary Grasses	7.1	6.1	8.2
Salix	Willow	3.9	2.0	5.8
Polygonum	Smartweeds	3.9	0.5	7.2
Chara	Muskgrass	3.8	7.7	0
Elodea	Waterweeds	3.4	2.7	4
Riccia	Liverworts	2.1	0	4.2
Stuckenia	Sago Pondweeds	1.7	3.4	0
Ceratophyllum	Coontails	1.6	2.4	0.7
Najas	Naiads	1.4	2.8	0
Sagittaria	Arrowhead	1.4	0.1	2.7
Utricularia	Bladderworts	1.3	0	2.6
Alisma	Water-plantains	1.1	0.3	1.9
Leersia	Cutgrasses	0.8	1.3	0.3
Veronica	Speedwell	0.7	0.5	0.9
Myriophyllum	Water Milfoil	0.6	1.3	0
Cyperus	Flatsedges	0.5	0.2	0.8
Bidens	Beggarticks	0.4	0.5	0.4
Carex	Sedges	0.4	0	0.8
Schoenoplectus	Bulrushes	0.4	0.7	0.03
Eleocharis	Spikerushes	0.2	0.4	0.02
Acer	Maples	0.2	0	0.3
Panicum	Panicgrass	0.02	0	0.03
Typha	Cattails	0.02	0	0.03

With ADONIS, we detected no difference in the macrophyte generic composition (Df = 1,10, F = 0.55, P = 0.81) between oxbow types. The NMDS results suggested that more macrophyte taxa (Fig. 2) were associated with lower turbidity ($r^2 = 0.74$, p = 0.003) and pH ($r^2 = 0.48$), and with finer substrate ($r^2 = 0.26$). Only a few taxa (*Bidens*, *Lemna*, and *Phalaris*) increased in abundance with greater turbidity.

Macroinvertebrate assemblage

We recorded 44 invertebrate taxa across two sampling years (Tables 4 and 6). Average invertebrate taxon richness was 18.6 taxa (range = 15.0–23.0) in non-tiled oxbows and 17.5 (range = 15.0–19.5) in multipurpose oxbows (ANOVA; F = 0.33, Df = 1, 10, p = 0.58). Oligochaeta was the most common taxon in both site types with a relative abundance of 21% in non-tiled oxbows, and 33% in multipurpose sites. We found an average numerical density of 13,098 (range = 2623–31,004) individuals per m^2 in non-tiled oxbows, and 15,650 (range = 52,207–35739) individuals

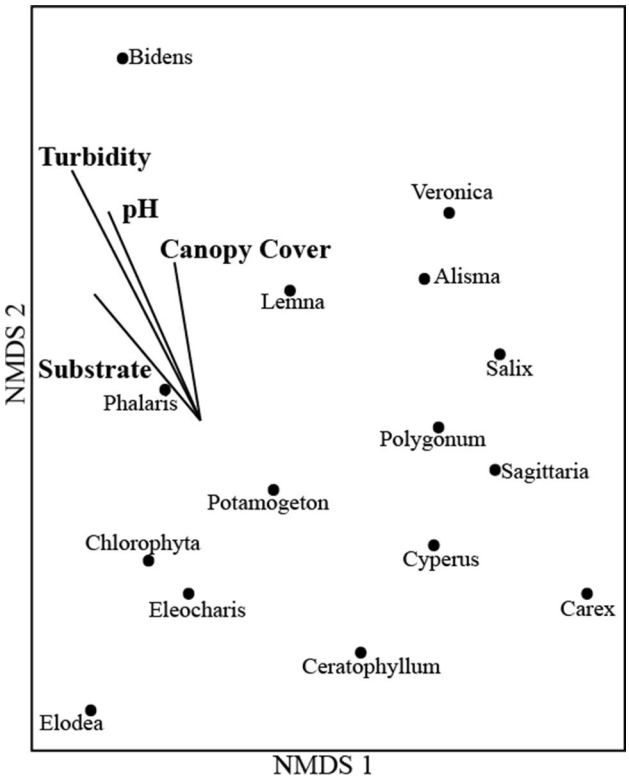


Fig. 2 NMDS ordination of the macrophyte assemblages of 12 oxbows averaged across two years. Genera found at more than two sites are present. Physical variables with an r^2 value ≥ 0.25 are represented as vectors

Table 6 Relative abundance of invertebrate taxa in oxbows wetlands (2019 and 2020)

		Relative Abundance (%)		
Taxa	Common name	Overall	Non-tiled oxbows	Multipurpose oxbows
Oligochaeta	Earthworms	27.6	21.04	33.1
Chironomidae	Non-biting midges	15.1	15.6	14.6
Cladocera	Water fleas	12.8	14.1	11.7
Copepoda	Copepod	9.9	11.2	12.3
Amphipoda	Scuds	7.6	9.7	4.6
Caenidae	Small squaregilled mayflies	5.8	7.1	2.5
Coenagrionidae	Narrow-winged damselflies	3.7	5.9	1.9
Planorbidae	Ramshorn snails	2.5	2.2	2.9
Physidae	Bladder snails	2.4	3.2	1.8
Ceratopogonidae	Biting midges	2.4	2.1	2.6
Nematoda	Roundworms	2.4	1.6	3.0
Sphaeriidae	Fingernail clams	1.6	0.3	2.6
Ostracoda	Seed shrimp	1.5	1.2	1.8
Corixidae	Water boatmen	0.8	0.6	1.0
Baetidae	Small minnow mayflies	0.6	0.4	0.7
Acari	-	0.6	0.8	0.3
Collembola	Springtails	0.5	0.03	0.9
Conchostraca	Clam shrimp	0.5	0.2	0.7
Libellulidae	Skimmer dragonflies	0.5	0.5	0.4
Hirudinea	Leeches	0.4	0.5	0.4
Haliplidae	Crawling water beetles	0.4	0.8	0.1
Mesoveliidae	Water treaders	0.1	0.2	0.1
Chaoboridae	Phantom midges	0.1	0.1	0.01
Hydroptilidae	Microcaddisflies	0.1	0.2	0
Tipulidae	Crane flies	0.1	0.02	0.1
Aeshnidae	Darner dragonflies	0.1	0.1	0.1
Culicidae	Mosquitoes	0.03	0.1	0.01
Pleidae	Pygmy backswimmers	0.03	0.1	0.01
Tabanidae	Horse and deer flies	0.03	0.1	0
Crambidae	Snout moths	0.02	0.1	0
Decapoda	Crayfish	0.02	0.02	0.02
Belostomatidae	Giant water bugs	0.02	0.04	0.001
Dysticidae	Predaceous diving beetles	0.02	0.02	0.01
Chrysomeliidae	Leaf beetles	0.01	0.03	0
Gyrinidae	Whirligig beetles	0.01	0.03	0
Leptoceridae	Long-horned caddisflies	0.01	0.02	0.001
Viviparidae	River snails	0.01	0.02	0
Lymnaeidae	Pond snails	0.002	0.001	0.003
		Relative Abundance (%)		
Taxa	Common name	Overall	Non-tiled oxbows	Multipurpose oxbows
Macromiidae	Cruiser dragonflies	0.002	0.004	0
Gomphidae	Club-tailed dragonflies	0.001	0.002	0.001
Unionidae	River mussels	0.001	0.001	0
Ephemeridae	Common burrower mayflies	0.001	0	0.001
Sciomyzidae	Marsh flies	0.001	0	0.001

per m^2 in multipurpose oxbows (ANOVA; F = 0.17, Df = 1, 10, p = 0.69).

We found no significant compositional difference in invertebrate assemblages between site types (ADONIS; Df = 1,10, F = 0.68, P = 0.63). The invertebrate NMDS (Fig. 3) ordination suggested that taxonomic composition was primarily affected by turbidity ($r^2 = 0.68$), macrophyte cover ($r^2 = 0.64$), depth ($r^2 = 0.32$), and pH ($r^2 = 0.26$). Few taxa (mainly decapods and chaoborid midges) were associated with higher turbidity. Similarly, decapods, chaoborid midges, and corixid bugs were found in greater numerical densities in higher pH sites. Planorbids and baetids were found in greater numerical densities in oxbows with a greater average depth. Many invertebrate taxa were associated with a higher macrophyte cover including pleid bugs, tipulid flies, and aeshnid dragonflies.

Fish assemblage

Over two field seasons, 199,325 individual fish representing 35 species were recorded (Table 7). We found an average total CPUE of 445.4 (range = 214.77–1055.26) in non-tiled oxbows and 362.92 in multipurpose oxbows (range = 24.1–799.39; F = 0.96, Df = 1, 10, p = 0.35). We found an average species richness of 8.2 (range = 1.5–17.0) in non-tiled oxbows and 11.4 (range = 4.0–14.5) in multipurpose oxbows (F = 1.10, Df = 1, 10, p = 0.32). Three fish species found in non-tiled oxbows were not found in multipurpose oxbows and two species in multipurpose oxbows were never recorded in non-tiled oxbows (Table 7). Fathead minnow (*Pimephales promelas*) was the most common species in both site types, with a relative abundance of 64.4% in non-tiled sites and 46.5% in multipurpose sites.

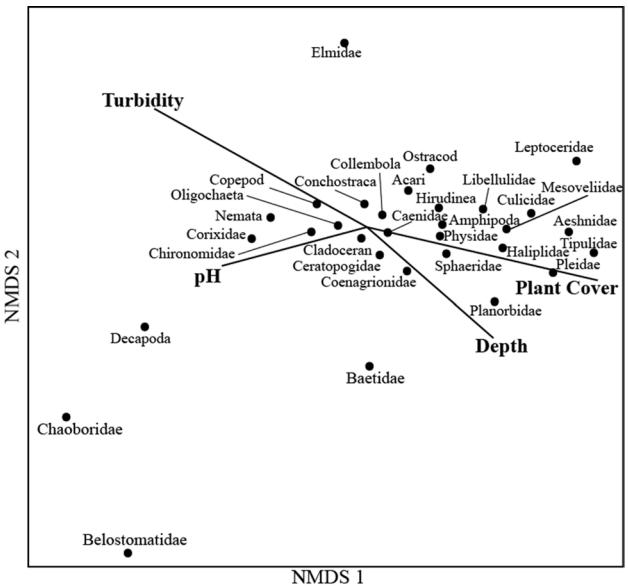


Fig. 3 NMDS ordination of the invertebrate assemblages of 12 oxbows averaged across two years. Taxa represented at more than two sites are present. Physical and biological variables with an r^2 value ≥ 0.25 are represented as vectors. To avoid overlap between taxa, lines are drawn to represent their places in ordination space

We found no significant compositional difference in fish assemblage between oxbow types (Df = 1,10, F = 1.25, P = 0.25). Fish NMDS (Fig. 4) ordination suggested that composition was primarily affected by plant cover ($r^2 = 0.77$, p = 0.002), invertebrate richness ($r^2 = 0.59$, p = 0.027), conductivity ($r^2 = 0.37$), pH ($r^2 = 0.28$), and tile ($r^2 = 0.27$). Oxbows with higher pH and temperature generally had greater abundance of bigmouth shiners (*Notropis dorsalis)*, sand shiners (*Notropis stramineus)*, western blacknose dace (*Rhinichthys obtusus)*, creek chubs (*Semotilus atromaculatus),* and bluntnose minnows (*Pimephales notatus)*. More turbid Oxbows with higher turbidity contained more brassy minnows (*Hybognathus hankinsoni)*, white suckers (*Catostomus commersonii)*, and common shiners (*Luxilis cornutus)*, while higher conductivity oxbows contained more largemouth bass (*Micropterus salmoides)* and black bullheads (*Ameiurus melas)*. Oxbows with higher macrophyte cover and macroinvertebrate richness supported greater abundances of brook sticklebacks (*Culaea inconstans*), fathead minnows, and green sunfish (*Lepomis cyanellus*).

Over two field seasons we found a total of 2682 Topeka shiners in three non-tiled and four multipurpose oxbows. We found an average Topeka shiner CPUE of 4.9 (range = 0–10.6 in non-tiled oxbows and 10.3 (range = 0–28.2) in multipurpose oxbows (Table 4). According to ADONIS, sites with Topeka shiner present did have a different fish composition (Df = 1, $r^2 = 0.26$, F = 3.561, p = 0.021), which SIMPER analysis suggested was driven primarily by common shiner, black bullhead, fathead minnow, and orangespotted sunfish (*Lepomis humilis*), together accounting for 50.6% of the fish assemblage variance between sites with and without Topeka shiner. There is some evidence that sites with Topeka shiner present had a differing invertebrate composition (Df = 1, $r^2 = 0.19$, F = 2.3, p = 0.066, α = 0.1), but SIMPER analysis suggested that this difference was spread out across many invertebrate taxa (planorbid snails, nematodes, amphipods, caenid mayflies, physid snails, baetid mayflies, ostracods, sphaeriids, libellulid dragonflies, corixid bugs, coenagrionid damselflies, and ceratopogonid midges accounted for 50.2% of variance).

Discussion

Compared to non-tiled oxbows in the agricultural landscape of north central and west central Iowa, we did not detect significant effects of tile drainage on physical variables and biological assemblages of multipurpose oxbows.

Table 7 Fish species recorded in all oxbow sites over 2019 and 2020 with relative abundance by taxa and listed in descending order of relative abundance

Taxa	Scientific name	Relative Abundance (%)		
		Overall	Non-tiled oxbows	Multi-purpose oxbows
Fathead minnow	*Pimephales promelas*	56.3	64.4	46.5
Black bullhead	*Ameiurus melas*	13.2	10.5	16.5
Green sunfish	*Lepomis cyanellus*	11.4	8.9	14.5
Common shiner	*Luxilis cornutus*	10.5	9.7	11.4
Orangespotted sunfish	*Lepomis humilis*	2.9	2.1	3.8
Topeka shiner	*Notropis topeka*	1.9	1.1	2.8
White sucker	*Catostomus commersonii*	1.0	0.7	1.3
Creek chub	*Semotilus atromaculatus*	0.6	0.7	0.5
Hornyhead chub	*Nocomis biguttatus*	0.5	0.4	0.7
Bluntnose minnow	*Pimephales notatus*	0.5	0.4	0.6
Brook stickleback	*Culaea inconstans*	0.3	0.1	0.6
Sand shiner	*Notropis stramineus*	0.3	0.4	0.1
Spotfin shiner	*Cyprinella spilotpera*	0.2	0.3	0.03
Bluegill	*Lepomis macrochirus*	0.1	0.1	0.1
Brassy minnow	*Hybognathus hankinsoni*	0.1	0.01	0.2
Largemouth bass	*Micropterus salmoides*	0.1	0.02	0.2
Golden redhorse	*Moxostoma erythrurum*	0.1	0.1	0.01
Western blacknose dace	*Rhinichthys obtusus*	0.1	0.1	0.1
Common carp	*Cyprinus carpio*	0.1	0.1	0.03
Bigmouth shiner	*Notropis dorsalis*	0.03	0.1	0.01
Carmine shiner	*Notropis percobromus*	0.01	0.02	0.01
Northern pike	*Esox lucius*	0.01	0.02	0.01
Johnny darter	*Etheostoma nigrum*	0.01	0.01	0.01
White crappie	*Pomoxis annularis*	0.01	0.02	0.004
Tadpole madtom	*Noturus gyrinus*	0.01	0.02	0.003
Central stoneroller	*Campostoma anomalum*	0.01	0.01	0.003
Yellow bullhead	*Ameiurus natalis*	0.01	0.01	0.004
Emerald shiner	*Notropis atherinoides*	0.01	0.01	0
Black crappie	*Pomoxis nigromaculatus*	0.004	0.01	0.003
Blackside darter	*Percina maculata*	0.003	0.003	0.002
Smallmouth bass	*Micropterus dolomieu*	0.002	0	0.003
Yellow perch	*Perca flavescens*	0.001	0.003	0
Bigmouth buffalo	*Ictiobus cyprinellus*	0.001	0.001	0
Shorthead redhorse	*Moxostoma macrolepidotum*	0.0003	0	0.001

While it is likely that contaminants from adjacent agricultural fields are introduced into all oxbows through groundwater and surface water pathways, recent evidence suggests that tile water is a significant contributor of nitrates in multipurpose oxbows (Pierce and Schilling 2023), and other work has found higher concentrations of selenium (Henry et al. 2020) and insecticides (Campbell et al. 2023) in tile-fed wetlands. Thus, the lack of an observed difference in oxbow fish and invertebrate assemblages should be considered preliminary but promising evidence for the efficacy of multipurpose oxbows in enhancing biodiversity in agricultural landscapes, restoring ecosystem services, and providing habitat for species of conservation concern (e.g., Topeka shiner).

This study represents one of the first efforts to document macroinvertebrate and macrophyte abundance and diversity of restored oxbows in the Midwest. Although oxbows likely exhibit flowing conditions when connected to adjacent streams, communities observed during our study were primarily typical of lentic ecosystems (Obolewski et al. 2016; Cavallaro et al. 2019). Across all sites, the most commonly occurring macrophytes were either macroscopic algae, floating, or floating-leaved, with a much smaller contribution of submerged aquatic vegetation, potentially suggesting

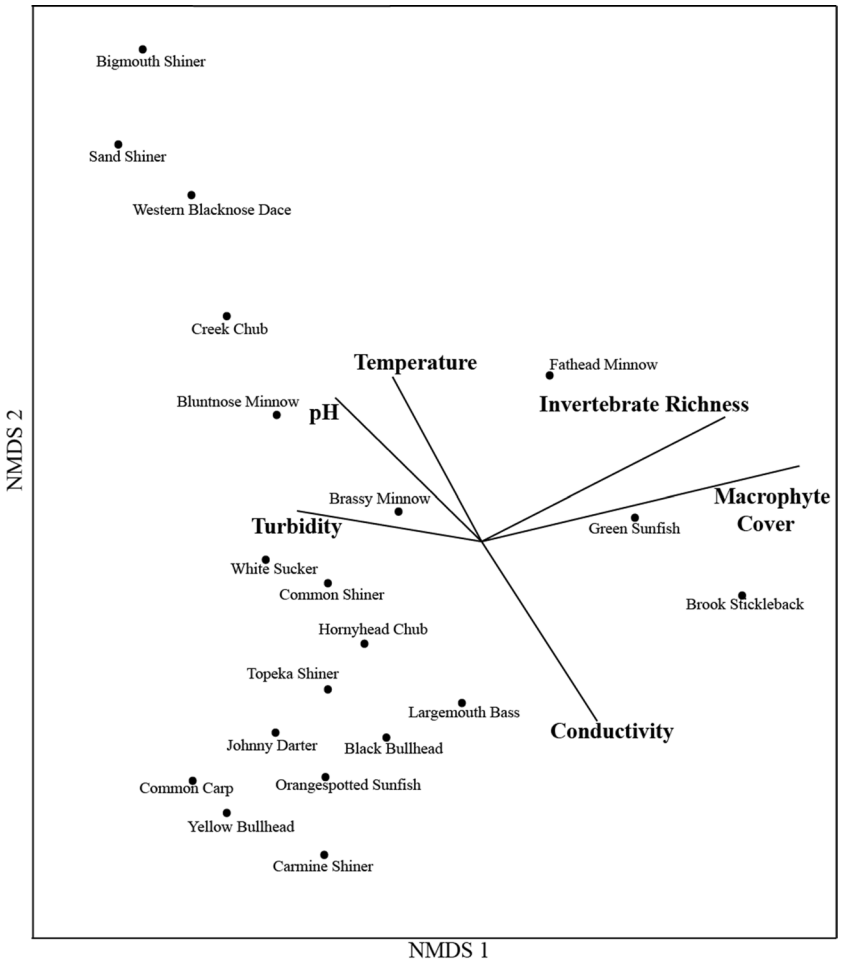


Fig. 4 NMDS ordination of the fish assemblages of 12 oxbows averaged across two years, Species represented at more than two sites are present. Physical and biological variables with an r^2 value ≥ 0.25 are represented as vectors

eutrophic conditions (Hansen et al. 2011). The five most common invertebrate taxa were the same across oxbow types, suggesting that additional nutrient contributions from subsurface drainage did not significantly alter macroinvertebrate assemblages and potentially points to eutrophic conditions (Bazzanti et al. 2012).

Oxbow community variation was primarily associated with turbidity. Low-turbidity communities exhibited high macrophyte cover and invertebrate richness. The few fish species present were green sunfish and fathead minnows, common in all oxbows, and brook stickleback which were more common in less turbid oxbows. Conversely, high-turbidity oxbows exhibited low macrophyte cover, few invertebrate taxa, and high fish richness (including Topeka shiner) and CPUE. Macroinvertebrates represented in these oxbows included corixid bugs, chaoborid midges, chironomid midges, decapods, and oligochaete worms. We found that variation was related across assemblages, with macrophyte variables predicting invertebrate assemblage measures and both macrophytes and macroinvertebrate variables related to fish variables. Topeka shiner presence varied with the abundances of other fishes and, to a lesser degree, invertebrates. The fish associates that we detected (common shiner, fathead minnow, orangespotted sunfish, and black bullhead) were similar to those found in other studies of Topeka shiner in oxbows (Simpson et al. 2019; Osterhaus et al. 2022b). However, as determined by Osterhaus et al. (2022b), associations between Topeka shiner and other fish species weaken with a larger sample size. Community variation may also be attributed to the frequency of oxbow connection to adjacent streams and the composition at time of disconnection (Osterhaus et al. 2022b). The ability of these restored wetlands to support diverse fish, macroinvertebrate, and macrophyte communities is encouraging, and contributes to the growing body of evidence that restored oxbows are effective in restoring the ecosystem services these once abundant habitats provide (Osterhaus et al. 2022b; Pierce and Schilling2023).

While we did not detect any differences in our biological variables between oxbow types, this study primarily examined differences in richness and abundance of invertebrates and fishes. This study did not assess impacts on morphology or behavior and was not designed to measure the sustained effects of individual contaminants associated

with tile water over time. Tile water has measurably higher concentrations of NO3-N compared to groundwater and non-tiled wetlands (Pierce and Schilling 2023). Nutrient pollution is known to reduce predation and foraging efficiency (Camargo and Alonso 2006; Nieman and Gray 2018). Additionally, tile water is typically associated with higher concentrations of pesticides and herbicides (Henry et al. 2020; Kraus et al. 2021; Campbell et al. 2023). Campbell et al. (2023) determined that higher neonicotinoid concentrations associated with tile water contributed to shorter cerebellum lengths and higher metabolite concentrations in juvenile leopard frogs. Furthermore, Henry et al. (2020) found that elevated selenium concentrations in tile water primarily contributed to mortality in adult life stages of invertebrates. Sustained exposure to the herbicides and pesticides often found in tile drainage can additionally contribute to less diverse macrophyte and invertebrate assemblages (Relyea 2005; Wang et al. 2021). Kraus et al. (2021) found lower biomass of emergent insects in wetlands receiving tile drainage, which may additionally expose terrestrial insectivores to contaminants.

Effects of contaminants can be exacerbated by increased duration of exposure, and thus are directly related to precipitation, flooding frequency and residence times (Gregoire et al. 2009; Li et al. 2021; Pierce and Schilling 2023). Each of the oxbows in this study was located closely to adjacent streams to facilitate frequent flooding and colonization for fish species, contributing to shorter residences times and potentially shorter exposure to contaminants (Fischer et al. 2018; Schilling et al. 2019). While residence times were not measured for all multipurpose oxbows in this study, an assessment of two of the study oxbows found residence times of 5.48 and 22.98 days in 2020–2021 (Pierce and Schilling 2023). It is worth noting that while 2020 was a drier year than 2019, and residence times were likely longer (Pierce and Schilling 2023), we found no consistent abundance or taxonomic difference between years.

Our study was limited to 12 oxbows of similar size and restoration date. As more multipurpose oxbows are implemented across the Midwest there will be greater opportunity to elucidate whether the lack of trends present in this study are an outlier or an accurate representation of the relationship of tiles with restored oxbow ecology. The potential difference in temperature between oxbow types is particularly worth further study. Significantly lower temperatures slow metabolic rate and can contribute to trophic cascades (Gray et al. 2014). Evidence also shows that colder wetlands are at somewhat less of a risk of algal blooms, but this could be countered by the additional introductions of nutrients from subsurface drainage (Jeppesen et al. 2000; Ontoria et al. 2019). Finally, the lack of an observed community difference between oxbow types may be attributed to high levels of background nutrients in surface and subsurface waters of an agricultural landscape.

This study provides preliminary evidence that impacts of tile drainage to the coarse physical characteristics of oxbows, and to the abundance and taxonomic structure of oxbow communities are negligible in a landscape dominated by agriculture. Recent research continues to establish the efficacy of restoring oxbows for conservation of Topeka shiners (Simpson et al. 2019; Pierce et al. 2019; Osterhaus et al. 2021), even documenting the return of the species to the White Fox Creek HUC10 after a three-decade absence (Osterhaus et al. 2021). Additionally, other recent research has provided evidence that restored oxbows fed by tile drainage reduce nutrient export to the stream (Schilling et al. 2019, 2020; Pierce and Schilling 2023). If further research affirms our findings, it can be reasonably assumed that the added influence of subsurface drainage does not meaningfully impact the biological community of restored oxbows in agricultural landscapes like Iowa and is likely beneficial to the recovery of Topeka shiner populations and restoration of ecosystem services. While restorations should proceed with careful monitoring, there is reason to be optimistic about the positive effect of continued implementation of multipurpose oxbows.

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Author Contributions All authors contributed to the formation and design of this study. Data collection, analysis, and interpretation were performed by Samuel Leberg and Dylan Osterhaus. Field contacts, permits, and grants were coordinated by Clay Pierce and Timothy Stewart. The first draft of the manuscript was written by Samuel Leberg. Dylan Osterhaus, Clay Pierce, and Tim Stewart commented on previous manuscript drafts. All authors read and approved the final manuscript.

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Data Availability The datasets that we generated during this study will be available on Dryad upon publication of this manuscript.

Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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